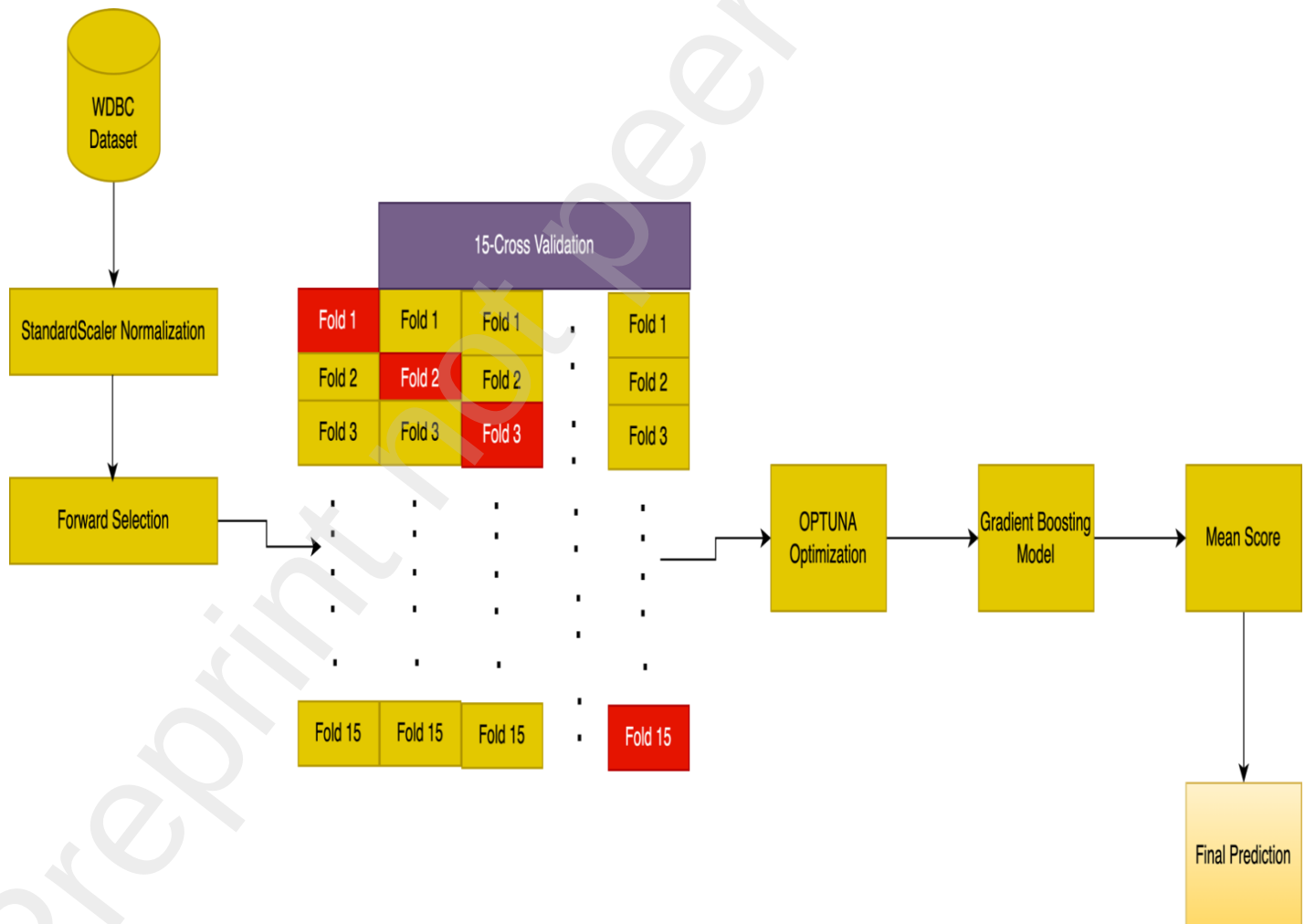


Graphical Abstract

Break the Ice (OPThyGB) : Hyper-Parameter Optimization Predictive Machine Learning Model framework for Early Warning Breast Cancer

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Overall operation of the proposed OPThyGB model system, which aims to predict breast cancer in women using Wisconsin breast Cancer Data set

Highlights

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Break the Ice (OPThyGB) : Hyper-Parameter Optimization Predictive Machine Learning Model framework for Early Warning Breast Cancer

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Abstract— Breast cancer, characterized by abnormal cell growth in the breast, predominantly affects women. In the pursuit of more accurate prediction and diagnosis, researchers are turning to artificial intelligence. One widely used dataset for this purpose is the Wisconsin Breast Cancer Dataset (WBCD), offering 30 metrics, including mean, standard error, and 'worst' values, to shed light on breast cancer characteristics. In this study, we introduce a novel model named OPThyGB, integrating Gradient Boosting with the OPTUNA optimizer. The model's performance is evaluated using essential metrics such as ROC curve, AUC, specificity, F1-score, sensitivity, and accuracy. Impressively, the hyperparameter optimization technique results in an outstanding 98.25% accuracy, surpassing benchmarks set by previous studies. This achievement not only showcases the model's exceptional accuracy but also positions it as a frontrunner in breast cancer prediction when compared to existing literature. The study underscores the potential of machine learning algorithms to significantly enhance breast cancer prediction and diagnosis, thereby contributing to improved clinical decision-making and ultimately better patient outcomes.

IndexTerms—Machine learning , OPTUNA , Optimization, Breast cancer , Hyper-parameter

1. INTRODUCTION

Ensuring breast health is a paramount concern for women aspiring to lead vibrant lives. However, the daunting specter of cancer poses a significant challenge, with Globocan estimating a staggering 2.3 million cases of breast cancer in women for the year 2022. The majority of benign tumors, fortunately, exhibit limited harm, refraining from infecting nearby organs or spreading. Their slow growth and well-defined boundaries often mitigate potential issues,

although challenges may arise depending on the tumor's size and location. For instance, a benign brain tumor could exert pressure on surrounding tissues, leading to symptoms. In contrast, malignant tumors pose a more serious threat, infiltrating adjacent tissues and organs, and potentially spreading through the bloodstream or lymphatic system. This invasive nature renders cancer particularly hazardous. Mutations in cells may disrupt the body's natural controls, allowing cancer cells to proliferate uncontrollably. Breast cancer, characterized by these two main types, underscores the critical necessity for early detection technology. The urgency lies in identifying the disease at its inception, before it has the chance to metastasize [1]. Incorporating modern technology and embracing cost-effective practices can alleviate the burden on healthcare providers and reduce overall healthcare expenditures. The widespread adoption of intelligent decision-making processes, coupled with advanced technology, holds the potential to save numerous lives. In the realm of machine learning model development, a crucial step involves hyperparameter tweaking. This process is integral to selecting optimal configurations for the training algorithm's hyperparameters, governing various aspects of the model such as learning rate, regularization, layer count, and node density [2]. The accuracy, generalization, and convergence

speed of a model hinge significantly on the thoughtful selection of hyperparameters. Optimal hyperparameter choices play a pivotal role in enabling the model to generalize well and accurately predict unfamiliar data. On the flip side, incorrect hyperparameters can lead to subpar performance, overfitting, or underfitting [3]. By optimizing hyperparameters, the machine learning model can be enhanced for improved accuracy, robustness, and efficiency [3].

While breast cancer is unavoidable, the prognosis and treatment costs significantly improve with early detection. Detecting breast cancer early can be challenging due to subtle symptoms. Therefore, engaging in regular mammography and self-breast examinations is crucial. These practices play a vital role in identifying abnormalities at an early stage, before tumors have a chance to grow [4].

1.1. *Objective and goal*

The study assesses currently available technologies and cost-effective methods to enhance early diagnosis and establish a dependable framework for detecting breast cancer. The aim is to elevate early diagnosis rates, reduce treatment expenses, and ultimately save lives through the implementation of machine learning (ML). In this research, our proposed framework will be evaluated against previous studies and ML models, utilizing accepted performance criteria. Our overarching objective is to advance breast cancer research and diagnosis by achieving these specific goals.

1.2. *Paper Layout*

The subsequent sections of the paper are structured as follows: In the second part, we conducted a thorough review of various research studies, delving into the effectiveness of algorithms for predicting breast cancer disease. Section 3 comprehensively explores numerous machine-learning methods. Moving forward, Section 4 details our recommended approach, encompassing model selection, parameter

configuration, experimental setup, and procedural aspects. The evaluation of performance measures is covered in Section 5, where we compare the proposed framework with existing ML models and literature. This section presents a comparative analysis of our findings against established models and literature. The paper concludes with Section 6, encapsulating final remarks and outlining potential future avenues for exploration.

2. RELATED WORK

In this section, we conducted an in-depth analysis of ten distinct research studies, providing insights into how scholars have approached the issue of the increasing number of breast cancer cases and their diverse methodologies. Researchers have been actively exploring various methods and models to enhance the accuracy of breast cancer prediction, resulting in numerous studies over the years. The following highlights key findings from these studies:

In their research endeavors, the authors have utilized a diverse array of algorithms, encompassing Bayesian network, Radial Basis Function, Back propagation network (BPN), Logistic Regression (LR), Artificial neural network (ANN), Convolutional neural network (CNN), k-nearest neighbours (KNN), Support vector machine (SVM), and Decision Tree Classifier (DT). The use of different machine learning algorithms, including KNN, Cubic SVM (CSVM), Simple Logistic Regression, Support Vector Machine (SVM), multilayer perceptron (MLP), and Optimized ANN, has been observed. Researchers typically start by applying the dataset on an individual basis, gaining insights into each algorithm's performance. Subsequently, the dataset is utilized to establish a comprehensive understanding. In one study [5], an examination of radiomics research on breast cancer axillary lymph node status yielded positive findings but also highlighted quality issues. The utilization of BOAALO, a hybrid feature selection technique, has been proposed to enhance breast cancer detection accuracy in another study [6], suggesting its potential application in diagnosis. A two-phase deep learning model coupled with random forest classifiers was introduced in [7], contributing to improved breast cancer prognosis using multi-

modal data. Notably, this model increased survival estimate sensitivity by 5.1%. Emphasizing the importance of early detection, a review paper [8] evaluated deep learning, machine learning, and complex data mining algorithms for breast cancer prediction in big data. In [9], researchers developed a DBT risk model, showcasing its potential to aid clinical decision-making and early detection efforts.

Positive correlations between fibroglandular volume (FGV) and breast cancer risk in Japanese women were revealed in [10], challenging the notion that

Table 1 . Evaluation of current methods for forecasting breast cancer.

Ref No.	Year	Algorithm	Dataset	Accuracy
[14]	2021	Bayesian network, Radial Basis Function	Wisconsin breast Cancer Data set (WBCD)	97%
[34]	2019	Back propagation network, ANN),(CNN), (SVM)	Wisconsin breast Cancer (original) dataset	94%
[35]	2023	An ensemble classifier based on multilayer perceptron.	Wisconsin Database	91%
[36]	2022	Heterogeneous ensemble model	Anthropometric data	78%
[17]	2017	RF-Recursive Feature Elimination (RF-RFE) method	Zhejiang Cancer Hospital	77.05%
[37]	2020	LR and DT	WDBC	94.4% and 95.1%
[38]	2018	NB and KNN	WBC	96.19% and 97.51%
[39]	2021	KNN	WDBC	96.8%,
[40]	2022	SVM, KNN,	Regional Oncology Center in Meknes, Morocco.	90.6%, 86.1%,
[41]	2022	KNN, SVM, LR and Random Forest Tree (RFT)	WDBC	91.25%, 92.5%, 93.75% and 95%
[42]	2023	Logistic Regression	CFS-BFS and Consistency-BFS	66%
[43]	2023	Classification performance was compared between Gail and BRISK.	Nested case-control dataset	(NRI = 0.287; SE = 0.035).

thick breast tissue may not be a substantial risk

Wisconsin Breast Cancer Data Collection

factor. In [11], authors presented a machine learning and image processing model for breast cancer diagnosis, incorporating picture analysis, feature extraction, selection, and various machine learning techniques. In the study [12], a comprehensive analysis of 91 current Explainable Artificial Intelligence (XAI) works spanning diverse domains aims to enhance the openness and reliability of AI systems. In a separate investigation [13], a meticulous examination delves into five machine learning algorithms, exploring their methodologies, real-world applications, and performance. Additionally, the research evaluates the potential of AI and ML across various sectors, highlighting the superiority of LSTM and SVM in specific contexts.

In their pursuit of a dependable breast cancer solution, the researchers scrutinized various criteria and employed classification matrices. The

(WBCD) served as the foundation for the utilization of Bayesian network and the Radial Basis Function method in [14]. In [15], SVM was applied to the Digital Database for Screening Mammography (DDSM). [16] employed KNN on the Federal Flu-minense University Hospital Mammography Dataset. Notably, RF-Recursive Feature Elimination (RF-RFE) achieved an accuracy of 77.05% when implemented on the Zhejiang Cancer Hospital dataset in [17].

2.1. Our main contribution

- We developed a powerful approach, dubbed OPThyGB, that detects breast cancer illness by using optimized GB and OPTUNA.
- GB, a tree-based distributed machine learning model, outperforms existing classification methods in speed and efficiency.
- We Utilise OPTUNA hyperparameter tuning which improved our system's performance.
- Evaluate the suggested model by contrasting its precision, sensitivity, and accuracy.

3. ALGORITHM USED & BACKGROUND

3.1 Algorithm used

In this section, we outline the machine learning classification methods employed in our research. To enhance accuracy, we optimized the classification model using the OPTUNA optimizer. We trained five different classifiers on the training dataset and selected the most efficient and robust one for hyper tuning.

A. K-Nearest Neighbour (KNN): KNN is a common non-parametric classification method. It classifies freshly found data using its k-nearest neighbors' classes and stores previously categorized data in a feature space. KNN is known for its simplicity and efficacy, although it may take longer to calculate big datasets, requires modifying

the number of neighbours (k), and is susceptible to outliers[18].

B. Random Forest (RF): Random Forest classifiers use decision trees, hierarchical structures based on dataset attributes. Random forests are multiple decision trees from bootstrap samples of the original dataset. They partition nodes using entropy or Gini. Suthaharan emphasizes bootstrapping to increase ensemble learning classification accuracy. Voting and bagging choose the finest trees. The recommended mental computing paradigm improves categorization using averaged random forests [19].

C. Decision Tree (DT): Decision tree classifiers are used in many applications because they can extract decision-making information from data. These basic data classification models create decision trees from training sets. AI rules these trees, categorizing data according to norms. Decision trees start classifying data points at the root node. The algorithm follows the branch and goes to the next node by comparing the original attribute with the attribute's values (the data point). The program repeats this process by comparing attribute values to additional children [20].

D. Gaussian Naive Bayes (GNB): Gaussian Naive Bayes classifies continuous data. It is particularly useful when class features are Gaussian (normal). For continuous data, the Gaussian distribution of continuous values for each class is a common assumption. The Gaussian Naive Bayes technique assumes normal distribution of continuous-valued dataset properties. It is helpful in many machine learning applications, particularly when the Gaussian distribution assumption is valid, since it can handle data with bell-shaped continuous variables [21].

E. Gradient Boosting (GB): Gradient boosting repeatedly fits a base learner to pseudo-residuals using least squares to improve regression models. The loss function gradients for model values at each data point during the current phase are pseudo-residuals. Randomness boosts accuracy and speed. Every iteration, a random subsample of the training dataset is taken without replacement. In the current iteration, this subsample substitutes

the whole dataset for fitting the base learner and calculating model updates, improving efficiency [42].

3.2 Scaling

Normalisation of the data is accomplished using the Standard scaler [44], which does this by eliminating the mean and scaling the data to unit variance. Mathematically speaking The equation that may be used to express the standard scaler is Equation 1, where 's' represents the standard deviation and 'μ' represents the mean.

$$x_{\text{scaler}} = \frac{x - \mu_{\text{mean}}}{s_{\text{stddiv}}} \quad (1)$$

3.3 Forward Selection

Forward selection in machine learning adds features to the model repeatedly depending on performance until a stopping requirement is fulfilled [22, 23]. Steps in the forward selection algorithm:

1. Start with an empty set of chosen characteristics S.
2. Train a model with features from S and calculate its performance for excluded features X.
3. Add the feature X with the greatest performance on the model trained with S union X to S.
4. Repeat steps 2 and 3 until a stopping condition is reached (e.g., a certain number of features are picked, performance improvement falls below a threshold, etc.).

3.4. K-Fold Cross-Validation

In the realm of machine learning, K-fold cross-validation stands as an approach for evaluating models when faced with limited data [24]. The process involves dividing the available data into k equal-sized "folds," wherein k-1 folds are utilized for training the model and the remaining one for validation. This iterative process is repeated k times, ensuring that each fold is employed once for validation while the others contribute to model training [25]. The steps in K-fold cross-validation can be outlined as follows:

1. Randomize the dataset.
2. Divide the dataset into k equal-sized groups.
3. Select each group as validation and the rest as training.
4. Use the training and validation sets to train and assess the model.
5. Calculate model evaluation metrics including accuracy, precision, recall, and f1-score.
6. Repeat steps 3-5 k times with a separate validation group.
7. Average the assessment measure across k repetitions.

K-fold cross-validation tests the model on multiple data subsets to limit the risk of overfitting. It also reduces evaluation metric variance by using several validation sets.

3.5. Optimization of OPTUNA

Machine learning models need hyperparameter adjustment to perform better. Creating good models requires this phase, which greatly affects model output. However, many people use trial and error to create hyperparameters and test them for hours or days. This method is inefficient and time-consuming. A program that automates hyperparameter tuning saves time and labor by removing manual testing. Four elements underpin OPTUNA's strategy to unite optimization paradigms.

3.5.1. Design-by-Run API

The OPTUNA design-by-run API optimizes hyperparameters during training, removing the need for separate optimization runs and improving efficiency. Design-by-run is used in deep learning to dynamically design deep networks for their intended purpose. Optimization uses this approach to dynamically generate the search space. Design-by-run lets us develop the search space depending on optimization objectives, making optimization more flexible and customizable [26]. OPTUNA describes hyperparameter optimization as optimizing an objective function by selecting the optimum hyperparameters to attain the highest validation score [27]. A trail object is used to generate the goal function, and the trail object's

functions dynamically create the search area as it runs. OPTUNA optimizes hyperparameters by repeatedly narrowing the search space based on objective function output.

3.5.2. Sampling

The hyperparameter optimization framework OPTUNA uses sampling. Relational and independent sampling are possible with OPTUNA [28]. Each hyperparameter is sampled independently. OPTUNA has Random, Grid, and Tree-structured Parzen Estimators. Hyperparameter correlations are considered in relational sampling. For relational sampling in high-dimensional and non-linear search spaces, OPTUNA leverages derivative-free optimization techniques like CMA-ES. Optimization task and search space dictate OPTUNA's sampling technique. For high-dimensional search spaces with intricate hyperparameter interactions, relational sampling methods like CMA-ES are better than independent sampling methods like TPE, which are computationally efficient. To test different algorithms and tailor optimization, OPTUNA enables users select unique sampling methodologies.

3.5.3. Pruning

Pruning decreases hyperparameter optimization computing [27]. OPTUNA has several pruning strategies for faster optimization. Successive Halving, an early stopping method, is a popular OPTUNA pruning algorithm [28]. The search space is divided into hyperparameter groups and evaluated concurrently. Remove the worst-performing half, then continue until one hyperparameter group remains [29]. Finding the optimal hyperparameters requires less trials. OPTUNA supports Asynchronous Successive Halving, which performs trials in parallel and dynamically allocates resources to promising trials and decreases resources for less promising ones. OPTUNA pruning algorithms reduce hyperparameter optimization computational cost and boost efficiency and effectiveness.

3.5.4 OPTUNA Easy -to -Simple Setup

OPTUNA's default memory data format makes it straightforward to use for fundamental tasks, which is important for a contemporary hyperparameter optimization framework. Advanced features including independent and relational sampling and trimming algorithms make OPTUNA a superior hyperparameter tuning framework. OPTUNA architecture is in Fig. 1. OPTUNA workers run their objective function (OF) once each sample. OPTUNA APIs access storage and retrieve relevant data from memory samples when the moment API is called [27]. Workers manage the OF independently and save research.

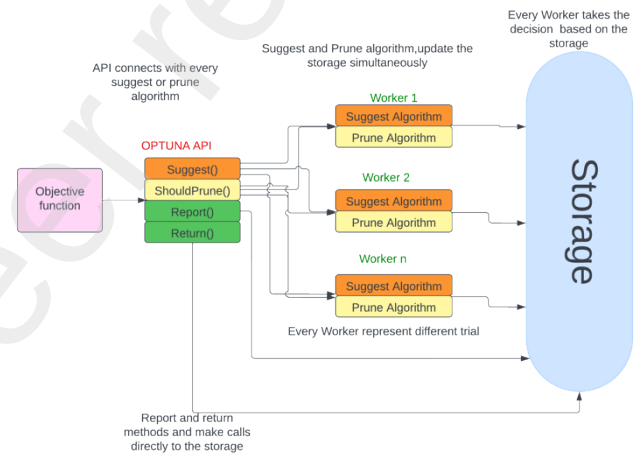


Fig 1. OPTUNA Architecture.

4. PROPOSED OPTHYGB FRAMEWORK

This section examined model selection criteria and parameters for framework-building algorithms. We summarized the experimental design and approach.

4.1. Model Selection

Fig. 2 shows the breast cancer prediction architecture. Section 4.3 describes how we utilized UCI Machine Learning Repository data [30]. We removed outliers after extensive data analysis and parameter correlations. Next, we split our data into training data (80%) and testing data (20%).

Our model was built and developed using the optuna optimization approach, which we then utilized to predict breast cancer using cross validation of several folds.

4.2. Tuning of OPTUNA Parameter

After scaling and feature selection, we optimized gradient boosting model hyperparameters using OPTUNA. We chose eight hyperparameters: loss, learning_rate, n_estimators, subsample, criteria, min_samples_split, leaf, and max_depth. After adjusting, we trained the gradient boosting model with optimal hyperparameters. Loss is the optimum loss function for binomial and multinomial deviation. Classification jobs with probabilistic outputs benefit from this function. Each tree's training effect is reduced by learning_rate. The number of gradient boosting rounds is determined by N_estimators. Gradient boosting resists overfitting, thus more rounds usually increase performance. Subsample is the percentage of training samples used to train each base learner. Criterion is the split effectiveness metric. The hyperparameter min_samples_split determines the minimal amount of data points in a decision tree node to split. Min_samples_leaf is a hyperparameter that sets the minimum number of instances in a tree leaf node. Maximum_depth hyperparameter limits a tree's depth by restricting nodes. Optimizing performance requires tuning this parameter, which relies on how input characteristics interact. If set to "None," the tree grows until all leaf nodes are pure or have less than "min_samples_split" samples. Optuna[27] is a framework that effectively identifies superior hyperparameters. We used Optuna to get the optimal hyperparameter values that maximize the fitness or objective function (*Fit*) score. The fitness function may be expressed as:

$$Fit = \left[\frac{1}{N_{CVT}} \sum_{i=1}^{N_{CVT}} F1_i \right] \quad (2)$$

$F1_i$ is the F1 score of the i th fold, and N_{CVT} represents the total number of cross-validation folds applied only to the training data. The research used the $N_{CVT} = 15$. A greater Fit value indicates the optimal value. By using a large amount of training data, we have set the N_{CVT} value to 15 to ensure that the suggested framework can prevent excessive latency, runtime, and

resource use, as opposed to when N_{CVT} is set to 20 or higher. The OPThyGB has not been used on the WDBC Datasets before, which presents a new application in the field of breast cancer detection.

Table 2. Default hyperparameters for gradient boosting model.

No	Parameters	Default
1	N_estimators	100
2	Subsample	1
3	Criterion	Friedman_mse
4	Learning_Rate	0.1
5	Min_samples_split	2
6	Max_depth	3
7	Min_samples_leaf	1
8	Max_features	Log2

Table 3. Optimized hyperparameters for gradient boosting model using OPTUNA.

No	Parameters	Default
1	N_estimators	550
2	Subsample	1
3	Criterion	Squared_error
4	Learning_Rate	0.5
5	Min_samples_split	2
7	Min_samples_leaf	1
8	Max_features	sqrt

4.3. Setup of Experiment

4.3.1. Data set

The Classification dataset Wisconsin-breast Cancer (Di-agnostics) (WBC) from the UCI machine learning repository measures breast malignant growth.

4.3.2. Visualization of Data and correlation

This section visualizes data, a crucial step in model development. Data visualization helps convey stories by simplifying data and showing patterns and outliers. In Fig.3. To compare the characteristics to the data, we used a heatmap.

Table 4 Nominal features descriptions.

Features	Description
ID number	It provides the specific identifier of a patient.
diagnosis	It is classified into two categories. M represents malignant and B represents benign.
perimeter	It specifies the boundary of the cell nucleus.
area	It delineates the region of the cell nucleus.
smoothness	The local variance lies in the lengths of the radii.
compactness	$(\text{perimeter}^2 / \text{area} - 1.0)$
concavity	(severity of concave segments of the contour)
symmetry	Consists of symmetry mean
radius	The mean of distances from the center to locations on the periphery is calculated

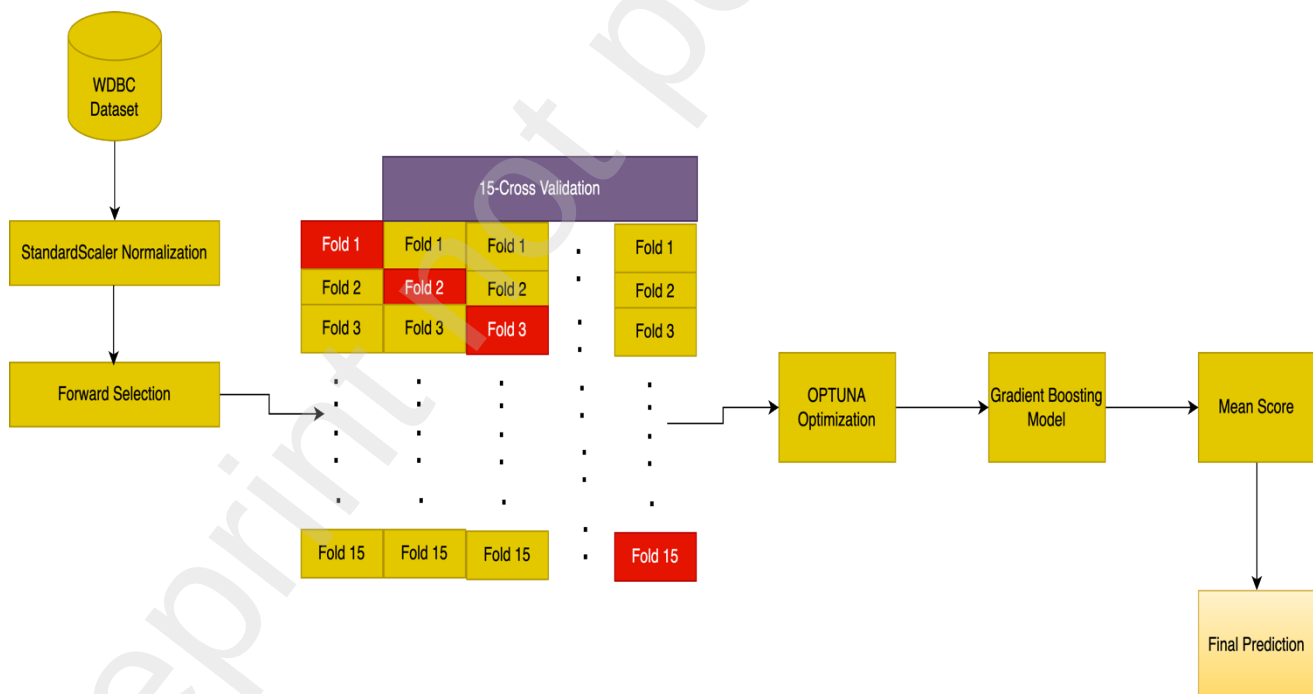
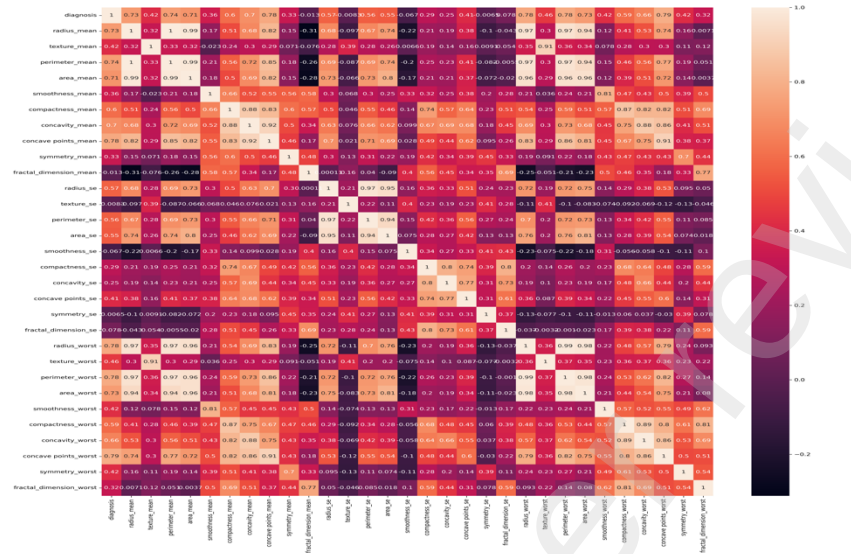


Fig 2. Proposed OPThyGB model framework for breast cancer detection.

Fig 3. Heatmap of



dataset features

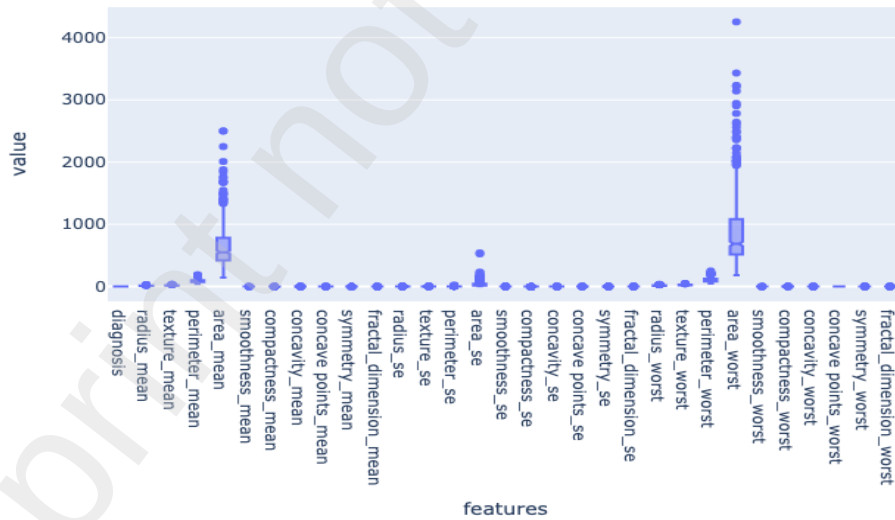


Fig 4. Features to value box plot of dataset

5. RESULTS & DISCUSSION

This section describes our framework's results and analysis. Several performance measures assessed the algorithms. We also compared our model to existing models for accuracy, precision, sensitivity, F1 Score, ROC, and MCC. Section 2 reviewed different algorithms and models' recommended model.

5.1. Performance metrics

Performance measurements include true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs).

TPR (sensitivity) [31]: It represents a classifier's possibility of predicting a positive result when the illness is present. The formula is:

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (2)$$

Specificity [31]: The chance of a classifier predicting a bad result without illness. Formula is:

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (3)$$

Accuracy [31]: Widely used to evaluate classifier performance. Calculated as a proportion of successfully recognized samples.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

AUC-ROC [32]: Area under the receiver operating characteristic curve. It is also a popular categorization performance statistic. Charts compare TPR and FPR at different thresholds. AU-ROC excels in performance comparison because it covers a broad variety of class distributions and error levels. The definition is:

$$AU - ROC = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (5)$$

F1 score [33]: Precision and memory weighted average (or harmonic mean). The greatest score is 1 and the poorest is 0. In F-measures, TNs are ignored. Formula for calculating F1 score as follows:

$$F1 \text{ Score} = 2 \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

5.2. Comparison with ML models

The use of an optuna optimizer hyper-tuning procedure, which results in improved accuracy and resilience, is what sets our method apart from other methods that have been undertaken in the past. Training our models on a substantial dataset and performing a comprehensive performance review using a variety of metrics are two things that we place a significant amount of importance on. The fact that our approach is able to combine high accuracy with efficient processing is a significant aspect that makes it particularly valuable for predicting breast cancer. Within the framework of our technique, the 15-fold cross-validation with optuna optimizer serves as the basis for determining which parameters for the gradient boost model are performing exceptionally well. In addition, the associated graph in Figure 5 demonstrates this precision. With a remarkable classification accuracy of 98.25%, the OPThyGB classifier surpasses other classifiers because of its superior performance. Different performance metrics, including accuracy, precision, sensitivity, and specificity, are used in order to conduct an in-depth comparison of our proposed framework with the existing machine learning models that are shown in Table 5. In addition, the area under the receiver operating characteristic curve (AUC-ROC) are shown in Figures 6,7,8,9,10 and 11 offer significant insight of confusion matrix of proposed model performance.

The levels of sensitivity and specificity that should be used are determined by the particular objectives of the model. In situations when it is essential to improve the exact categorization of positive instances, sensitivity is given emphasis by the organization. However, when the primary objective is to reduce the number of instances that

are incorrectly classified as negative, specificity is the most important factor to consider. Remembering that changes in threshold may have an effect on both sensitivity and specificity is a very important point to keep in mind. When assessing the performance of a machine learning model, it is advantageous to offer both metrics since they provide a more comprehensive view of the model's efficiency. As a consequence, it is ideal to supply both measurements simultaneously.

Table 5
Evaluation of proposed framework with existing ML models.

Model	Accuracy	Precision	Sensitivity	Specificity	F1-Score
GNB	92.11%	92.50%	86.05%	95.77%	89.16%
DT	85.09%	76.00%	88.37%	83.10%	81.72%
KNN	93.86%	95.00%	88.37%	97.18%	91.57%
RF	94.74%	93.02%	93.02%	95.77%	93.02%
Proposed OPThyGB	98.25%	98.04%	98.04%	98.41%	98.04%

Table 6. OPThyGB mode performancel using different k-fold cross-validation.

No of Trials	K-Fold Cross validation	Accuracy
100	K =5	93.83%
100	K = 10	95.56%
100	K = 15	98.25%

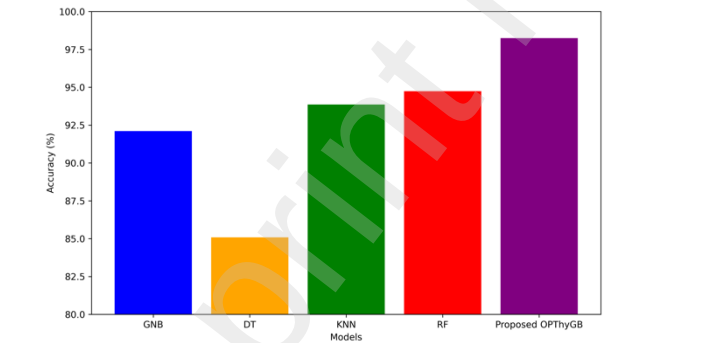


Fig. 5. Model classifiers best prediction accuracy

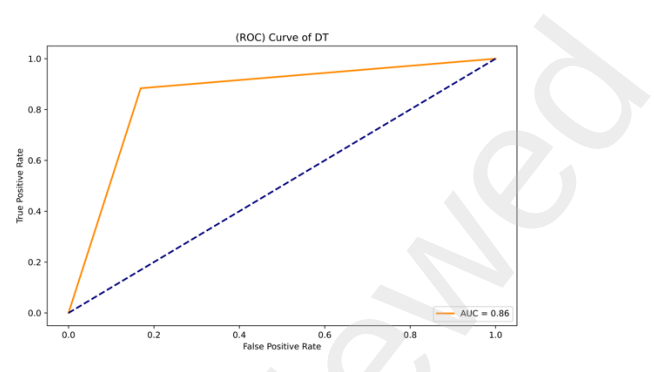


Fig. 6. ROC curve for proposed framework

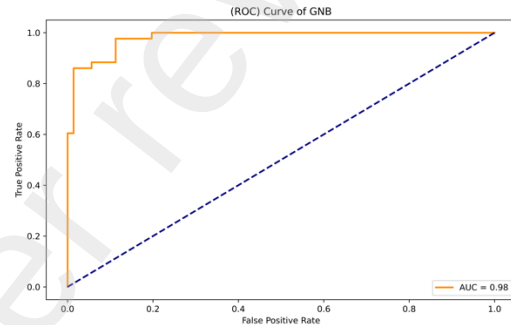


Fig. 7. ROC curve for proposed framework

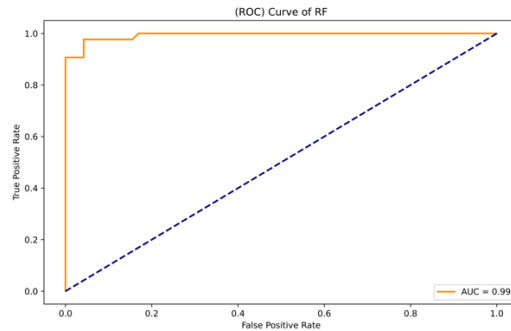


Fig. 8. ROC curve for proposed framework

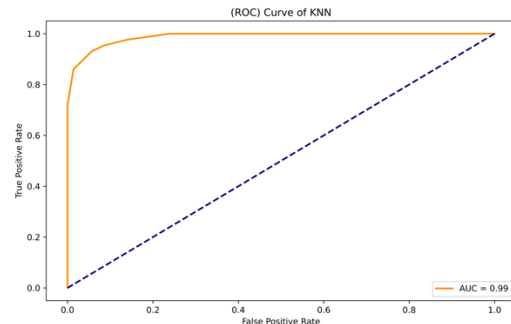


Fig. 9. ROC curve for proposed framework

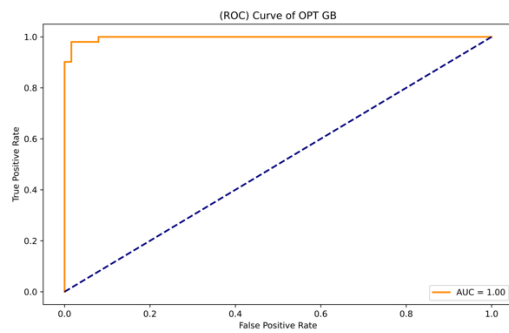


Fig. 10. ROC curve for proposed framework

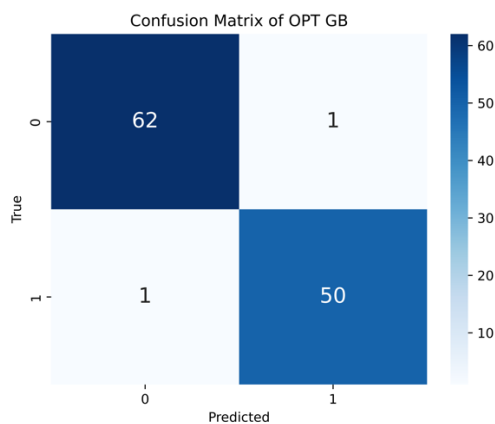


Fig. 11. Confusion Matrix for proposed framework

5.3. Comparison with existing literature

In this section, we conduct an in-depth analysis of how our proposed technique compares to the corpus of knowledge that has been available before. The similarities and differences of the algorithms that are being considered are what distinguish this comparison from others. An informative depiction of the outcomes of the study can be seen in Figure 12, which provides information that may be of significant use to medical research and diagnostics. The medical community is provided with a more in-depth understanding of these algorithms in the context of breast cancer research via the use of our technique, which provides a clear portrayal of the many features and performance patterns of each algorithm.

Despite the fact that a great number of

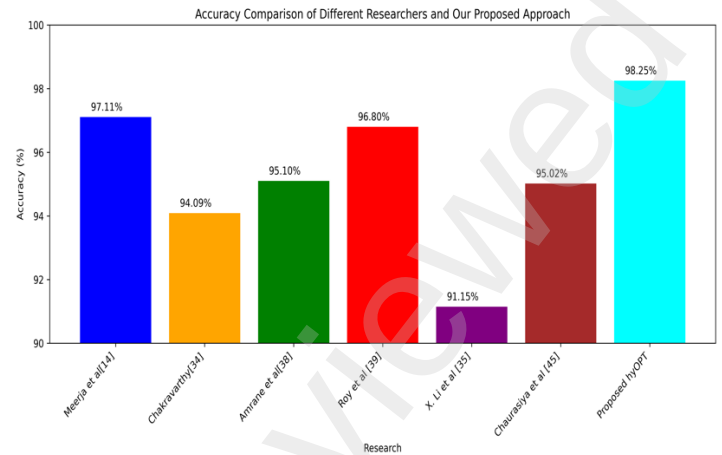


Fig. 12. Different accuracies achieved by different researchers and our proposed approach

researchers have explored a variety of algorithms and datasets for the purpose of breast cancer detection, our model stands out owing to the exceptional performance it has. In comparison to other models, ours is distinguished by its ability to achieve the highest possible level of accuracy.

V. CONCLUSION

The suggested framework would significantly benefit regions in West Africa such as Ghana, Nigeria, Benin and Liberia, where the population is growing and healthcare resources are limited. Early detection of illnesses, particularly breast cancer, is crucial in the current challenging healthcare setting. This framework offers the possibility for early detection, benefiting patients, and considerably enhances the healthcare sector's capacity to identify cancer at an early stage. This approach has shown resilience and surpasses current research by using stacked machine learning methods, as indicated in the results section.

There is much room for growth and progress in the future. In the future, the technique will be evaluated on bigger datasets to enhance its applicability and reliability. Moreover, integrating deep learning is expected to enhance the system's accuracy and diagnostic capabilities further. Another intriguing avenue for improvement is to expand the array of algorithms used inside the framework. This technique has the potential to enhance the

framework's effectiveness in early breast cancer detection, leading to improved healthcare outcomes and perhaps saving lives.

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Declarations

Conflict of interest

The authors declare that they have no conflict of interest.

Ethics approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Data availability

The datasets are available on reasonable request.

REFERENCES

- [1] Mangukiya, M., Vaghani, A., & Savani, M. (2022). Breast cancer detection with machine learning. *International Journal for Research in Applied Science and Engineering Technology*, 10(2), 141-145.
- [2] Schratz, P., Muenchow, J., Iturritxa, E., Richter, J. and Brenning, A., 2019. Hyperparameter tuning and performance assessment of statistical and machine-learning algorithms using spatial data. *Ecological Modelling*, 406, pp.109-120.
- [3] Yang, L. and Shami, A., 2020. On hyperparameter optimization of machine learning algorithms: Theory and practice. *Neurocomputing*, 415, pp.295-316.
- [4] Anji Reddy Vaka, B. Soni, K. Sudheer Reddy, Breast cancer detection by leveraging machine learning, *ICT Express* 6 (4) (2020) 320–324, <http://dx.doi.org/10.1016/j.icte.2020.04.009>.
- [5] A. Calabrese, D. Santucci, R. Landi, Bruno Beomonte Zobel, E. Faiella, Carlo de Felice, Radiomics MRI for lymph node status prediction in breast cancer patients: the state of art, *J. Cancer Res. Clin. Oncol.* 147 (6) (2021) 1587–1597, <http://dx.doi.org/10.1007/s00432-021-03606-6>.
- [6] Shankar Thawkar, S.J. Sharma, M. Khanna, Law Kumar Singh, Breast cancer prediction using a hybrid method based on butterfly optimization algorithm and ant lion optimizer, *Comput. Biol. Med.* 139 (2021) 104968, <http://dx.doi.org/10.1016/j.combiomed.2021.104968>.
- [7] N. Arya, S. Saha, Multi-modal advanced deep learning architectures for breast cancer survival prediction, *Knowl. Based Syst.* 221 (2021) 106965, <http://dx.doi.org/10.1016/j.knosys.2021.106965>.
- [8] Madhu Kirola, Minakshi Memoria, A. Dumka, K. Joshi, A comprehensive review study on: Optimized data mining, machine learning and deep learning techniques for breast cancer prediction in big data context, *Biomed. Pharmacol. J.* 15 (1) (2022) 13–25, [Online]. Available: <https://biomedpharmajournal.org/vol15no1/a-comprehensive-review-study-on-optimized-data-mining-machine-learning-and-deep-learning-techniques-for-breast-cancer-prediction-in-big-data-context/>. (Accessed: Oct. 26, 2023).
- [9] A risk model for digital breast tomosynthesis to predict breast cancer and guide clinical care, *Sci. Transl. Med.* (2022) <https://www.science.org/doi/full/10.1126/scitranslmed.abn3971>. (Accessed Oct. 26, 2023).
- [10] M. Nara, T. Fujioka, M. Mori, Tomoyuki Aruga, U. Tateishi, Prediction of breast cancer risk by automated volumetric breast density measurement, *Jpn. J. Radiol.* (2022) <http://dx.doi.org/10.1007/s11604-022-01320-y>.
- [11] J. Prasad, et al., Computational technique based on machine learning and image processing for medical image analysis of breast cancer diagnosis, *Secur. Commun. Netw.* 2022 (2022) 1–7, <http://dx.doi.org/10.1155/2022/1918379>.
- [12] A. Saranya, R. Subhashini, A systematic review of explainable artificial intelligence models and applications: Recent developments and future trends, *Dec. Anal. J.* 7 (2023) 100230, <http://dx.doi.org/10.1016/j.dajour.2023.100230>.
- [13] M. Bansal, A. Goyal, A. Choudhary, A comparative analysis of K-nearest neighbor, genetic, support vector machine, decision tree, and long short term memory algorithms in machine learning, *Dec. Anal. J.* 3 (2022) 100071, <http://dx.doi.org/10.1016/j.dajour.2022.100071>.
- [14] Meerja Akhil Jabbar, Breast Cancer Data Classification using Ensemble Machine Learning, *Engineering and Applied Science Research*, vol. 48, (1) 2021, pp. 65– 72, [Online]. Available: <https://ph01.tci-thaijo.org/index.php/easr/article/view/234959>. (Accessed: Oct. 25, 2023).
- [15] S. Ge, et al., Variability measurement for breast cancer classification of mam-mographic masses, in: *Dntb.Gov.Ua*, 2015, <https://ouci.dntb.gov.ua/en/works/4MEKy3b4/>. (Accessed Oct. 25, 2023).
- [16] T.M. Mejia, María Jesús Verdú Pérez, V.H. Andaluz, A. Conci, Automatic Segmentation and Analysis of Thermograms

using Texture Descriptors for Breast Cancer Detection, 2015, <http://dx.doi.org/10.1109/apcase.2015.12>.

[17] Z. Jiang, Classification of benign and malignant breast cancer based on DWI texture features, in: Proceedings of the International Conference on Bioinformatics and Computational Intelligence, in: ACM Other Conferences, 2017, <https://dl.acm.org/doi/abs/10.1145/3135954.3135964>. (Accessed Oct. 25, 2023).

[18] M. Forina, M. Casale, P. Oliveri, Application of Chemometrics To Food Chemistry, Elsevier eBooks, 2009, pp. 75–128, <http://dx.doi.org/10.1016/b978-044452701-1.00124-1>.

[19] P.D. Caie, N. Dimitriou, Ognjen Arandjelović, Precision Medicine in Digital Pathology Via Image Analysis and Machine Learning, Elsevier eBooks, 2021, pp. 149–173, <http://dx.doi.org/10.1016/b978-0-323-67538-3.00008-7>.

[20] Henrique Luiz Corrêa, Agile Manufacturing as the 21st Century Strategy for Improving Manufacturing Competitiveness, 2001, <http://dx.doi.org/10.1016/b978-008043567-1/50001-2>.

[21] P. Majumder, Gaussian naive Bayes, in: OpenGenus IQ: Computing Expertise & Legacy, 2020, <https://iq.opengenus.org/gaussian-naive-bayes/>. (Accessed Oct. 25, 2023).

[22] Meyer, H., Reudenbach, C., Hengl, T., Katurji, M. and Nauss, T., 2018. Improving performance of spatio-temporal machine learning models using forward feature selection and target-oriented validation. *Environmental Modelling & Software*, 101, pp.1-9.

[23] Dietterich, T.G., 2002. Machine learning for sequential data: A review. In *Structural, Syntactic, and Statistical Pattern Recognition: Joint IAPR International Workshops SSPR 2002 and SPR 2002 Windsor, Ontario, Canada, August 6–9, 2002 Proceedings* (pp. 15-30). Springer Berlin Heidelberg.

[24] Alkhamash, E.H., Hadjouni, M. and Elshewey, A.M., 2022. A Hybrid Ensemble Stacking Model for Gender Voice Recognition Approach. *Electronics*, 11(11), p.1750.

[25] Wong, T.T. and Yeh, P.Y., 2019. Reliable accuracy estimates from k-fold cross validation. *IEEE Transactions on Knowledge and Data Engineering*, 32(8), pp.1586-1594.

[26] Agrawal, T. and Agrawal, T., 2021. Optuna and autoML. *Hyperparameter Optimization in Machine Learning: Make Your Machine Learning and Deep Learning Models More Efficient*, pp.109-129.

[27] Akiba, T., Sano, S., Yanase, T., Ohta, T. and Koyama, M., 2019, July. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 2623-2631).

[28] Lai, J.P., Lin, Y.L., Lin, H.C., Shih, C.Y., Wang, Y.P. and Pai, P.F., 2023. Tree-Based Machine Learning Models with Optuna in Predicting Impedance Values for Circuit Analysis. *Micromachines*, 14(2), p.265.

[29] Tonmoy, S.T.I. and Zaman, S.M., 2022, December. OOG-Optuna Optimized GAN Sampling Technique for Tabular Imbalanced Malware Data. In *2022 IEEE International Conference on Big Data (Big Data)* (pp. 6534-6539). IEEE.

[30] UCI machine learning repository, in: Uci.Edu, 2019, <https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>. (Accessed Oct. 25, 2023).

[31] A. Baratloo, M. Hosseini, A. Negida, G. El Ashal, Part 1: Simple Definition and Calculation of Accuracy, Sensitivity and Specificity, Emergency (Tehran, Iran), vol. 3, (2) 2015, pp. 48–49, [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4614595/>. (Accessed: Oct. 25, 2023).

[32] J. Brownlee, How to use ROC curves and precision-recall curves for classification in python MachineLearningMastery.com, in: MachineLearningMastery.Com, 2018, [https://machinelearningmastery.com/roc-curves-and-precision-recall-curves-for-classification-in-python/#:~:text=with%20sample%20code,-,ROC%20Curves%20and%20AUC%20in%20Python,probabilities%20for%20the%201%20class.\(accessed%20Oct.%2025,2023\).](https://machinelearningmastery.com/roc-curves-and-precision-recall-curves-for-classification-in-python/#:~:text=with%20sample%20code,-,ROC%20Curves%20and%20AUC%20in%20Python,probabilities%20for%20the%201%20class.(accessed%20Oct.%2025,2023).)

[33] Joos Korstanje, The F1 score, in: Towards Data Science, 2021, Medium. Published August 31, 2021. Accessed May 11, 2022. <https://towardsdatascience.com/the-f1-score-bec2bbc38aa6>.

[34] T. Chakravarthy, Analysis the Breast Cancer using Back Propagation with Deep Neural Network, vol. 7, (4) 2019, pp. 844–847, [Online]. Available: https://www.ijcseonline.org/full_paper_view.php?paper_id=4128. (Accessed: Oct. 25, 2023).

[35] X. Li, X. Chen, Amin Rezaeipannah, Automatic breast cancer diagnosis based on hybrid dimensionality reduction technique and ensemble classification, J. Cancer Res. Clin. Oncol. 149 (10) (2023) 7609–7627, <http://dx.doi.org/10.1007/s00432-023-04699-x>.

[36] S. Nanglia, M. Ahmad, F. Khan, N.Z. Jhanjhi, An enhanced predictive heterogeneous ensemble model for breast cancer prediction, Biomed. Signal Process. Control 72 (2022) 103279, <http://dx.doi.org/10.1016/j.bspc.2021.103279>.

[37] Sengar, P.P., Gaikwad, M.J. and Nagdive, A.S. (2020) Comparative Study of Machine Learning Algorithms for Breast Cancer Prediction. Proceedings of 2020 Third

International Conference on Smart Systems and Inventive Technology (ICSSIT), Tirunelveli, 20-22 August 2020, 796-801. <https://doi.org/10.1109/ICSSIT48917.2020.9214267>
<https://ieeexplore.ieee.org/abstract/document/9214267>

[38] Amrane, M., Oukid, S., Gagaoua, I. and Ensari, T. (2018) Breast Cancer Classification Using Machine Learning. Proceedings of 2018 Electric Electronics, Computer Science, Biomedical Engineerings' Meeting (EBBT), Istanbul, 18-19 April 2018, 1-4.
<https://ieeexplore.ieee.org/abstract/document/8391453>
<https://doi.org/10.1109/EBBT.2018.8391453>

[39] Roy, S., Gawande, R., Nawghare, A. and Mistry, S. (2021) Comparative Study of Machine Learning Algorithms for Detecting Breast Cancer. International Journal of Computer Science Trends and Technology (IJCST), 9, 103-111.

[40] Merouane, E. and Said, A. (2022) Prediction of Metastatic Relapse in Breast Cancer using Machine Learning Classifiers. International Journal of Advanced Computer Science and Applications, 13, 176-181.
<https://doi.org/10.14569/IJACSA.2022.0130222>

[41] Chaurasiya, S. and Rajak, R. (2022) Comparative Analysis of Machine Learning Algorithms in Breast Cancer Classification. (Preprint) <https://doi.org/10.21203/rs.3.rs-1772158/v1>

[42] M. Lamba, G. Munjal, Yogita Gigras, M. Kumar, Breast cancer prediction and categorization in the molecular era of histologic grade, Multimedia Tools Appl. 82 (19) (2023) 29629–29648, <http://dx.doi.org/10.1007/s11042-023-14918-9>.

[43] R. Allman, Y. Mu, G.S. Dite, E.L. Spaeth, J.L. Hopper, B. Rosner, Validation of a breast cancer risk prediction model based on the key risk factors: family history, mammographic density and polygenic risk, Breast Cancer Res. Treat. 198 (2) (2023) 335–347, <http://dx.doi.org/10.1007/s10549-022-06834-7>.

[44] J. Hale. *Scale, Standardize, or Normalize With Scikit-Learn*. (2019) Accessed: Apr. 8, 2021. [Online]. Available: <https://towardsdatascience.com/scale-standardize-or-normalize-with-scikit-learn-6ccc7d176a02>.